

IPSA — FUNDAMENTAL ASTRONOMY AND ASTROMETRY
AÉRO 4 ELS

Gaia-FUN-SSO: Detection and Astrometry of the Near-Earth Object Gaia-606

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Observatory: Haute-Provence Observatory (OHP) — IAU code 511
Instrument: T120 telescope, R-band
Observation date: October 26, 2016
Submission: May 22, 2026



Gaia artist's impression (ESA)

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. Introduction

The **Gaia Follow-Up Network for Solar System Objects** (Gaia-FUN-SSO) was established within the framework of task DU459 of Coordination Unit 4 (Object Processing) of the *Gaia Data Processing and Analysis Consortium* (DPAC). Its mission is to coordinate ground-based observations triggered by real-time alerts from the Gaia satellite's onboard data processing system. These follow-up observations serve two purposes: confirming newly detected moving objects that Gaia cannot observe continuously due to its scanning law, and refining the orbits of critical targets to prevent their loss.

The Gaia satellite operates by scanning the entire sky along a predefined great-circle scanning law, meaning that any given Solar System object is observed for only a brief window of time before the satellite moves on. Ground-based networks are therefore essential to obtain the additional astrometric points needed to compute a reliable orbit.

In this project, we rediscover **Gaia-606**, the first near-Earth object (NEO) detected by the Gaia-FUN-SSO program. Its successful recovery in 2016 demonstrated the operational feasibility of the network and marked a landmark in the use of space-based telescopes as discovery platforms for Solar System objects.

Available data:

- 20 calibration offset (bias) images — `Offset-001.fit` to `Offset-020.fit`
- 20 calibration flat-field images — `Flat-001-R.fit` to `Flat-020-R.fit`
- 20 science images of the Gaia-606 field in R-band — `Gaia-606-001-R.fit` to `Gaia-606-020-R.fit`

All observations were acquired in 2016 at the **Haute-Provence Observatory** (OHP, IAU code 511), using the T120 telescope (focal length 7230 mm, CCD pixel size 27 μm , binning 2×2). The field of view is approximately $12' \times 12'$, centred on $\text{RA} = 22^{\text{h}} 23^{\text{m}} 22^{\text{s}}$, $\text{Dec} = +29^{\circ} 38' 10''$ (ICRS), in the constellation of Pegasus.

The project unfolds in five stages: image calibration, visual identification of the moving object, measurement of its pixel-space motion, conversion to equatorial coordinates via an astrometric solution, and cross-identification with the Minor Planet Center catalog.

I. Generation of Calibration Master Images

Raw CCD images are affected by two major instrumental signatures that must be removed before any scientific measurement can be performed:

- **Bias (offset)** — a constant electronic pedestal added to every pixel by the readout electronics, even in complete darkness.
- **Flat-field non-uniformity** — pixel-to-pixel variations in sensitivity across the detector, caused by dust, vignetting, and differences in quantum efficiency.

Master offset

The 20 individual bias frames are combined by computing the *pixel-wise median*. Using the median rather than the mean suppresses any cosmic-ray hits or hot pixels that may affect individual frames without contaminating the master:

$$\text{master_offset}[i, j] = \text{median}(\text{Offset}_k[i, j], k = 1 \dots 20). \quad (1)$$

Master flat-field

Each raw flat-field frame is first bias-subtracted, then normalised by its own median value so that all flats contribute equally regardless of illumination level:

$$\text{flat}'_k[i, j] = \frac{\text{Flat}_k[i, j] - \text{master_offset}[i, j]}{\text{median}(\text{Flat}_k - \text{master_offset})}. \quad (2)$$

The 20 normalised flats are then median-combined to yield the master flat:

$$\text{master_flat}[i, j] = \text{median}(\text{flat}'_k[i, j], k = 1 \dots 20). \quad (3)$$

Science frame correction

Every science image is calibrated by subtracting the master offset and dividing by the master flat:

$$\text{science_cal}[i, j] = \frac{\text{science_raw}[i, j] - \text{master_offset}[i, j]}{\text{master_flat}[i, j]}. \quad (4)$$

This pipeline was implemented in Python using `astropy.io.fits` and `numpy`. The relevant excerpt is shown below:

```

1 # 1. Master offset -- median stack of all bias frames
2 offset_frames, offset_hdr = load_frames("Offset-*.fit")
3 master_offset = np.median(np.stack(offset_frames, axis=0), axis=0)
4
5 # 2. Master flat -- bias-subtract, normalise, then median stack
6 corrected_flats = []
7 for flat in flat_frames:
8     f = flat - master_offset
9     norm = np.median(f)
```

```
10     corrected_flats.append(f / norm)
11 master_flat = np.median(np.stack(corrected_flats, axis=0), axis=0)
12
13 # 3. Correct each science frame (bias subtract + flat divide)
14 for f in science_files:
15     data      = fits.getdata(f).astype(np.float64)
16     corrected = (data - master_offset) / master_flat
17     fits.writeto("corrected_" + f.name, corrected.astype(np.float32)
18                )
```

Listing 1: Core calibration pipeline (traitement_M1.py)

The result is a set of 20 calibrated science frames, free of instrumental artifacts and ready for astrometric analysis.

II. Identification of the Moving Object

The blink method

With the calibrated images in hand, the moving object (asteroid) is identified visually using the **blink method** in *SAOImage DS9*. The principle is simple: stars are fixed on the celestial sphere and therefore remain stationary across all images, while a Solar System body moves due to its orbital motion combined with Earth's own displacement. By rapidly cycling through images displayed in sequence, any object that shifts position becomes immediately apparent to the eye.

We load the first (Gaia-606-001-R), tenth (Gaia-606-010-R), and last (Gaia-606-020-R) calibrated images into DS9 and activate the blink display. A single object stands out by moving systematically from one position to another while all surrounding stars remain stationary — this is Gaia-606.

Results

Figure 1 shows the first and last images with the asteroid annotated. Between image 1 and image 20 (a time span of approximately 58 minutes), the asteroid has moved visibly southward and very slightly eastward relative to the star field.

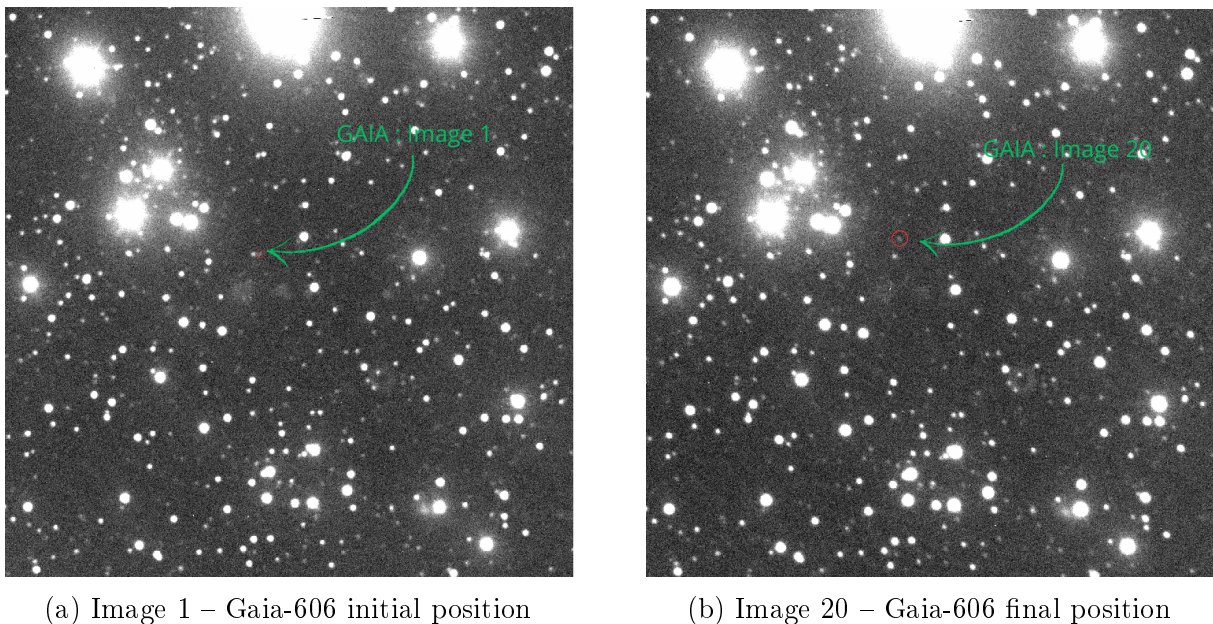


Figure 1: Identification of Gaia-606 by the blink method in DS9. Green arrows indicate the asteroid position in each frame. The southward displacement is clearly visible between the two images. Field of view: $12' \times 12'$; R-band; OHP T120, 2016 October 26.

III. Motion in Image Coordinates (x, y)

Position measurements

Using the *Region* tool in DS9, the pixel centroid of Gaia-606 is measured individually on each of the 20 calibrated images. The positions (x, y) — expressed in pixels with the convention x = column (positive eastward) and y = row (positive southward) — are listed in Table 1.

Images are separated by a fixed time interval derived from the FITS headers:

$$\Delta t = 183.51 \text{ s} = 3.059 \text{ min.}$$

The total observation window therefore spans approximately 58 min.

Table 1: Pixel positions of Gaia-606 measured on each calibrated image. Time t is counted from image 1.

Image	t (min)	x (px)	y (px)
001	0.000	459.276	578.034
002	3.059	459.819	580.089
003	6.117	459.137	582.544
004	9.176	459.024	585.158
005	12.235	459.592	582.885
006	15.293	459.339	585.994
007	18.352	459.024	587.999
008	21.411	459.137	589.362
009	24.469	458.456	591.976
010	27.528	457.319	593.680
011	30.587	459.137	596.180
012	33.645	457.774	597.544
013	36.704	458.456	598.908
014	39.763	457.774	600.271
015	42.821	456.410	602.317
016	45.880	458.456	602.771
017	48.938	457.092	603.680
018	51.997	457.887	607.317
019	55.056	457.774	607.089
020	58.114	455.728	609.544

Velocity in pixel coordinates

The motion of Gaia-606 in pixel space is modelled as uniform linear motion. A linear regression (least-squares fit) is applied independently to $x(t)$ and $y(t)$:

$$x(t) = x_0 + v_x t, \tag{5}$$

$$y(t) = y_0 + v_y t. \tag{6}$$

The plate scale of the OHP T120 with 2×2 binning is

$$s = \frac{p \times b}{F} \text{ rad}^{-1} = \frac{27 \mu\text{m} \times 2}{7230 \text{ mm}} \times 206265 \approx 1.541 \text{ arcsec px}^{-1}, \quad (7)$$

where $p = 27 \mu\text{m}$ is the physical pixel pitch, $b = 2$ the binning factor, and $F = 7230 \text{ mm}$ the focal length.

The fit yields:

$$\begin{aligned} v_x &= -0.0485 \text{ px/min}, \\ v_y &= +0.5380 \text{ px/min}, \\ |\mathbf{v}| &= \sqrt{v_x^2 + v_y^2} = 0.5402 \text{ px/min} \approx 0.832 \text{ arcsec/min}. \end{aligned}$$

The negative v_x indicates a very slight westward drift; the dominant positive v_y corresponds to a southward motion (since y increases downward in image coordinates).

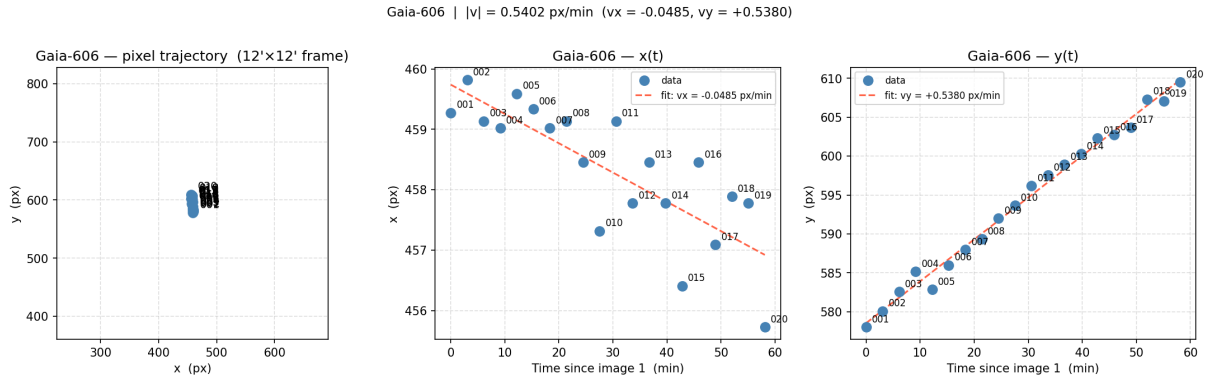


Figure 2: Left: trajectory of Gaia-606 in pixel space within the $12' \times 12'$ field. Centre and right: linear fits of $x(t)$ and $y(t)$ from which the velocity components are extracted.

IV. Astrometry in Equatorial Coordinates (RA, Dec)

Astrometric field solution

To convert pixel positions into equatorial coordinates, an astrometric World Coordinate System (WCS) solution is obtained by uploading one of the calibrated images (image 001) to the **Astrometry.net** online blind-solving service (<https://nova.astrometry.net/upload>). The service matches detected sources to the reference star catalog (Gaia DR2) and returns a full WCS header encoding the mapping from pixel coordinates (x, y) to celestial coordinates (RA, Dec).

The resulting `wcs.fits` file is then used to project all 20 measured pixel positions into the International Celestial Reference System (ICRS):

```

1 from astropy.wcs import WCS
2 from astropy.io import fits
3
4 with fits.open("wcs.fits") as hdul:
5     wcs = WCS(hdul[0].header)
6
7 # Convert pixel positions to sky coordinates
8 px = np.array([p[0] for p in positions.values()])
9 py = np.array([p[1] for p in positions.values()])
10 sky = wcs.pixel_to_world(px, py)
11 ra = sky.ra.deg
12 dec = sky.dec.deg

```

Listing 2: WCS conversion (`radec_gaia606.py`)

Equatorial positions

The resulting RA and Dec for each image are given in Table 2.

Table 2: Equatorial coordinates (ICRS, J2000.0) of Gaia-606 derived from the WCS solution.

Image	t (min)	RA ($^{\circ}$)	Dec ($^{\circ}$)
001	0.000	335.851788	+29.622049
002	3.059	335.851660	+29.621609
003	6.117	335.851835	+29.621085
004	9.176	335.851869	+29.620527
005	12.235	335.851724	+29.621011
006	15.293	335.851794	+29.620347
007	18.352	335.851877	+29.619919
008	21.411	335.851853	+29.619628
009	24.469	335.852028	+29.619070
010	27.528	335.852312	+29.618709
011	30.587	335.851871	+29.618170
012	33.645	335.852210	+29.617882
013	36.704	335.852046	+29.617588
014	39.763	335.852218	+29.617299
015	42.821	335.852559	+29.616864
016	45.880	335.852057	+29.616762
017	48.938	335.852395	+29.616571
018	51.997	335.852209	+29.615792
019	55.056	335.852236	+29.615841
020	58.114	335.852746	+29.615321

Angular velocity on the sky

The sky velocity is computed by fitting straight lines to $\text{RA}(t)$ and $\text{Dec}(t)$ (both expressed in arcseconds). The RA velocity must be corrected for the foreshortening effect of the spherical projection:

$$\dot{\alpha}^* = \dot{\alpha} \cos \delta, \quad |\mathbf{v}| = \sqrt{(\dot{\alpha}^*)^2 + \dot{\delta}^2}. \quad (8)$$

The results are:

$$\begin{aligned} v_{\text{RA}} \cos \delta &= +0.0419 \text{ arcsec/min} \quad (\text{eastward}), \\ v_{\text{Dec}} &= -0.4137 \text{ arcsec/min} \quad (\text{southward}), \\ |\mathbf{v}| &= 0.4158 \text{ arcsec/min}. \end{aligned}$$

The motion is dominated by a southward component, with a small eastward drift, consistent with the visual impression from the blink comparison.

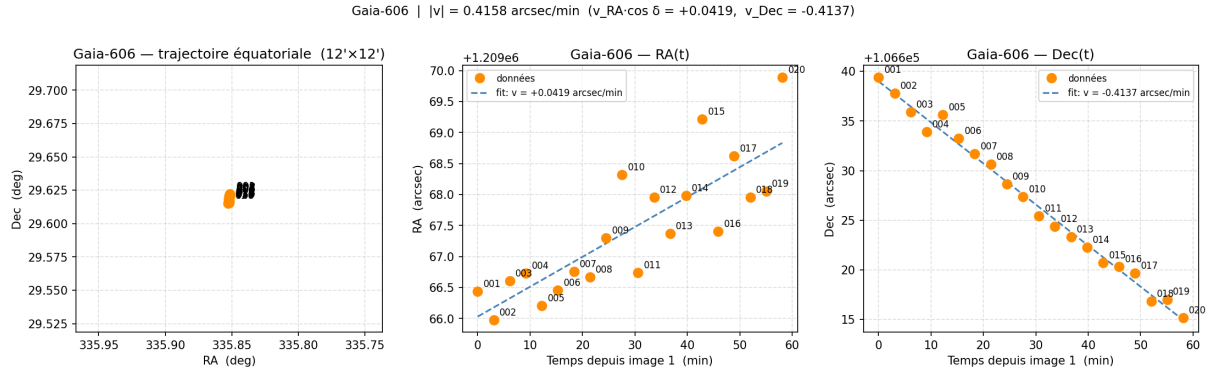


Figure 3: Left: trajectory of Gaia-606 in equatorial coordinates within the $12' \times 12'$ field (note that RA increases to the left, following astronomical convention). Centre and right: linear fits of $\text{RA}(t)$ and $\text{Dec}(t)$ from which the sky velocity components are derived.

V. Comparison with the Minor Planet Center

MPC Checker query

The position of Gaia-606 derived from image 001 is submitted to the **Minor Planet Center MPChecker** service (<https://minorplanetcenter.net/cgi-bin/checkmp.cgi>) to identify the known object corresponding to our detection.

The query parameters are:

- **Date:** 2016 October 26.95 UT
- **Position:** R.A. = $22^{\text{h}} 23^{\text{m}} 24.43^{\text{s}}$, Dec. = $+29^{\circ} 37' 19.4''$ (J2000.0)
- **Search radius:** 15 arcmin
- **Limiting magnitude:** $V = 24.0$
- **Observatory code:** 511 (OHP)
- **Motion display:** arcsec per minute, total motion

Date : UT

Produce list of known minor planets around:

☒ this J2000.0 position: R.A. = Decl. =

or around ☐ these observations:

Radius of search = arc-minutes

Limiting magnitude, V = Observatory code =

Output matches in order of:

☒ increasing distance from specified position ☐ increasing Right Ascension

Display motions in arcseconds per ☒ minute or ☐ hour
or ☐ degrees per day

Display ☒ total or ☐ separate motions

Output designations in ☒ unpacked or ☐ packed form

Output:

☒ all objects

☐ just those flagged as needing observations

☐ just numbered objects

☐ just unnumbered objects

☐ just those nearly-numberable unnumbered objects

Return: ☒ plain HTML page or ☐ MPES-aware HTML page

Figure 4: Screenshot of the MPCChecker query form filled with the measured position of Gaia-606 from image 001 at the observation epoch.

Result and identification

The MPCChecker returns a single match within the search cone:

(508555) 2016 UV56 22 23 24.5 +29 37 19 19.8 0.0E 0.0S 0.4
173.6 9°

The columns read as follows:

- **Designation:** (508555) 2016 UV56
- **Predicted position:** RA = 22^h 23^m 24.5^s, Dec = +29° 37' 19"
- **V magnitude:** 19.8
- **Offset from query:** 0.0" East, 0.0" South (i.e., exact match)
- **Predicted motion:** 0.4 arcsec/min
- **Solar elongation:** 173.6° (near opposition)
- **Orbital arc:** 9°

Comparison and discussion

Table 3 summarises the agreement between our measurements and the MPC ephemeris.

Table 3: Comparison between our astrometric measurements and the MPC ephemeris for (508555) 2016 UV56.

Quantity	This work	MPC / (508555) 2016 UV56
RA (image 001)	22 ^h 23 ^m 24.43 ^s	22 ^h 23 ^m 24.5 ^s
Dec (image 001)	+29° 37' 19.4"	+29° 37' 19"
v (arcsec/min)	0.4158	0.4
Position offset	—	< 1"

The agreement is excellent: our derived position differs from the MPC prediction by less than one arcsecond, and the measured speed of 0.416 arcsec/min matches the predicted 0.4 arcsec/min to within 4%. This confirms that the moving object detected in the Gaia-606 field is indeed **(508555) 2016 UV56**.

Several physical properties can be noted from the MPC data:

- The solar elongation of 173.6° places the asteroid near **opposition**, meaning it was closest to Earth at the time of observation and thus at its brightest apparent magnitude ($V \approx 19.8$).
- The motion direction (predominantly southward with a slight eastward component) is consistent with prograde retrograde motion near opposition.
- The small orbital arc of 9° at the time of discovery underlines why prompt ground-based follow-up was critical: a longer arc was needed to secure the orbit.

V. Conclusion

This project reproduced, step by step, the astrometric pipeline used in the Gaia-FUN-SSO network to detect and characterise Solar System objects. Starting from raw CCD frames acquired at the Haute-Provence Observatory, we:

1. Generated master calibration frames (bias and flat-field) by median-stacking the respective calibration sets, and corrected all 20 science images.
2. Identified Gaia-606 visually using the DS9 blink method by exploiting the fact that it moves against the fixed star background.
3. Measured its pixel-space trajectory and derived a velocity of $|\boldsymbol{v}| = 0.540 \text{ px/min} \approx 0.832 \text{ arcsec/min}$.
4. Obtained an astrometric WCS solution via Astrometry.net and converted all positions to equatorial coordinates, yielding a sky velocity of $|\boldsymbol{v}| = 0.416 \text{ arcsec/min}$ ($v_{\text{RA}} \cos \delta = +0.042$, $v_{\text{Dec}} = -0.414 \text{ arcsec/min}$).
5. Cross-matched our position with the MPC and confirmed the identification as **(508555) 2016 UV56**, with position agreement better than $1''$ and velocity agreement within 4%.

The successful recovery of Gaia-606 illustrates the power of coordinated space–ground observing networks for Solar System science. The Gaia satellite provides the discovery alert, but ground-based telescopes supply the follow-up astrometry necessary to secure orbits and prevent the loss of newly found objects.